

GAWTAM CHITHRA RAMESH

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Education

KTH Royal Institute of Technology

Master of Science in Systems, Controls, and Robotics (GPA 4.62/5.0)

Aug 2024 – Jun 2026

Stockholm, Sweden

Indian Institute of Technology Madras

Dual Degree (B.Tech Engineering Design and M.Tech Robotics) (GPA 7.9/10)

Aug 2019 – Jul 2024

Chennai, India

- Awarded (among 30 of 800+) the IIT Madras **Young Research Fellowship** based on remarkable research potential.
- Secured **97.64** Percentile in JEE Advanced and **99.10** Percentile in JEE Mains among **1.5M** candidates.
- Secured an All India Rank of **52** out of 50k in Undergraduate Common Entrance Examination for Design.

Skills : Python, C++, C#, PyTorch, JAX, CUDA, Slurm, ROS2, MuJoCo, PyBullet, OpenCV, wandb, Docker, Git

Professional Experience

Master's Thesis: Generative Flow Matching for Robot Manipulation

Jan 2026 – Jun 2026

ABB Robotics

Västerås, Sweden

- Designed a generative flow-matching trajectory planner guided at inference by Signal Temporal Logic gradients.
- Steered a single pretrained flow model toward unseen temporal-logic specifications at inference without retraining.
- Benchmarked on long-horizon reach-avoid navigation against state-of-the-art specification-guided generative planners.

Device Research Intern: 3D Scene Graphs

Jul 2025 – Dec 2025

Ericsson Research

Lund, Sweden

- Integrated instance segmentation and tracking into Hydra, a real-time 3D scene-graph construction framework.
- Built a synchronized RGB-D inpainting pipeline that removes privacy-sensitive objects from scene graphs online.
- Benchmarked state-of-the-art generative inpainting models to balance reconstruction fidelity against real-time latency.

Multi-Object Detection & Tracking Intern

Jun 2021 – Jul 2021

Blurgs Research Labs

Remote, India

- Evaluated five state-of-the-art multi-object tracking architectures for an Indian Navy video-surveillance product.
- Trained deep detection-and-tracking models on aerial drone and CCTV footage for real-time multi-target tracking.
- Deployed the top model for online tracking and re-identification of multiple moving targets across video streams.

Research Experience

Taxonomy-Guided Vision-Language Models for Manipulation

Jan 2025 – Nov 2025

Prof. Florian T. Pokorny, Robotics, Perception & Learning Lab, KTH

Stockholm, Sweden

- Coupled vision-language foundation models with a taxonomy to interpret deformable-object manipulation scenes.
- Built a prompt-taxonomy framework mapping natural language to robot-parsable structured motion primitives.
- Quantified taxonomy-grounding and natural-language correctness with F1-score and text-similarity metrics.

Autonomous Ultrasound Scan via Structure-from-Motion

Jan 2024 – May 2024

Prof. Nirav Patel, Department of Engineering Design, IIT Madras

Chennai, India

- Reconstructed a dense 3D torso point cloud from uncalibrated monocular RGB images via structure-from-motion.
- Densified the reconstruction with multi-view stereo and published the surface as a ROS point-cloud message.
- Segmented the abdominal region and estimated surface normals to plan the probe approach in a ROS/MoveIt pipeline.

Technical Projects

Deep Learning Advanced | DD2610, KTH

Nov 2025 – Jan 2026

- Reimplemented a one-step generative flow-matching model in PyTorch, ablating its time-sampling schedule and ratios.
- Showed accuracy-based generative quality metrics break down under classifier-free guidance mixing at sampling time.
- Built a Masked Autoencoder ViT from scratch with an asymmetric encoder-decoder and 75% random patch masking.
- Pretrained a contrastive self-supervised encoder, reaching 86% linear accuracy on CIFAR-10 using only 1% labels.

3D Open-Vocabulary Semantic Mapping | DD2600 Robot Learning, KTH

Aug 2025 – Oct 2025

- Built a CLIP-based semantic voxel-mapping pipeline from RGB-D scans for open-vocabulary object-presence detection.
- Fused open-vocabulary detection with promptable segmentation to ground object instances in a 3D voxel map.
- Trained a ResNet18 presence classifier from scratch, reaching 86.2% validation accuracy and 0.917 ROC-AUC.

Artificial Intelligence & Multi-Agent Systems | DD2438, KTH

Feb 2025 – May 2025

- Coordinated 20+ holonomic and non-holonomic agents collision-free using conflict-based multi-agent path-finding.
- Trained three role-specific PPO policies for attack, defend, and escape in a multi-agent capture-the-flag game.
- Fused a behavior tree with reinforcement learning via self-play and curriculum training across staged episodes.

Lead Author Publications

- Robust Object-layer Construction of 3DSG Using Instance Segmentation (IMPROVE26)(**Best Paper Award**)
- Synchronized RGB-D Inpainting for Privacy-Aware 3D Scene Graph Construction.(**ICPR26**).(Under Review)
- Scene Understanding in Deformable Object Manipulation via Taxonomy-Guided VLMs.(**IROS25 Workshop**)